Type MS · Explain diffusion across a partially permeable membrane in dialysis and Visking tubing

Answer 1 of 7 · Question 8 part b.i · 4 marks

2020 JAN · Paper 01 · find part b.i in this mark scheme extract

Number				
8	(a) (i)	X = urethra (1) Y = ureter (1)		2
	(ii)	Any two from: • transports oxygenated blood from the heart (1) • to the body/named organ (1) • maintains high blood pressure (1)		2
	(iii)	Any two from • blood in A/renal artery contains more oxygen/oxygenated (1) • blood in A/renal artery contains more urea (1) • blood in A/renal artery contains more ions (1) • similar concentration of glucose/B renal vein has less glucose (1) • blood in A/renal artery contains less carbon dioxide (1)	Allow reverse argument for B/renal vein	2
	(b) (i)	Any four from: • dialysis fluid/dialysate contains a low(er) concentration/no urea/waste products/ions (1) • urea/waste/ions diffuse from blood (1) • through dialysis membrane (1) • from high(er) concentration (in the blood) to a low(er) concentration (in the dialysate)/down a concentration gradient (1) • dialysis membrane is partially-permeable(1)	ORA allow selectively permeable membrane	4
	(ii)	• to prevent blood from clotting whilst in the machine (1) • (added as it enters the machine) to prevent clots from blocking the machine (as blood flows through) (1) • (not added at the end) as blood clotting properties need to return to normal before passing back into patient (1)		3

Total for question 8 = 13 marks

Type MS · Explain diffusion across a partially permeable membrane in dialysis and Visking tubing

Answer 2 of 7 · Question 6 part b · 3 marks

2021 JUN · Paper 01 · extract 1 of 2 · find part b in this mark scheme extract

number				
6 (a)	(i)	• separating different sized molecules out of the blood/only allows small molecules to pass; • under pressure;		2
	(ii)	• A has more glucose; • A has more urea; • A more amino acids;	ORA vessel B allow references to more oxygen in A and less carbon dioxide in A	3
	(iii)	• Vessel A bring blood to Bowman's capsule/glomerulus; • vessel B has narrow(er) lumen/diameter; • causes increase/higher blood pressure; • to force out molecules;		4
(b)		• glucose would pass from blood; • respiration affected/reduced; • lack of energy/ATP;		3

Total 12 marks

Type MS · Explain diffusion across a partially permeable membrane in dialysis and Visking tubing

Answer 2 of 7 · Question 6 part b · 3 marks

2021 JUN · Paper 01 · extract 2 of 2 · find part b in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain diffusion across a partially permeable membrane in dialysis and Visking tubing

Answer 3 of 7 · Question 5 part b · 3 marks

2025 JUN · Paper 01 (regional R) · extract 1 of 2 · find part b in this mark scheme extract

5 (a)	(i)	• flows from artery; • where blood is at high pressure; • into vein; • because blood is at a low pressure;	mp1- accept in context of artery leaving heart mp3- accept in context vein of returning to the heart If comparative answer given for pressure score marking point 2 and 4	4
	(ii)	glomerulus;		1
	(iii)	• large (surface) area; • for diffusion / molecules/ named molecules to pass out(of blood);	More area for diffusion = 2 marks	2
	(iv)	more urea;		1
(b)		• no/less gradient between liquid X and blood (glucose)/blood and X have same concentration/eq; • glucose does not pass out/ glucose is not lost: • by diffusion • needed for respiration; • minimises osmotic effect;		3
(c)		• protein digested/broken down into <u>amino acids</u>; • (excess) **<u>amino acids</u>** broken down/deaminated; • urea; • less protein means less urea produced; • less urea means less to be filtered/removed ; • high concentration of urea/nitrogenous waste can be toxic;	MP3– accept urea from protein breakdown	4
				Total =15

Type MS · Explain diffusion across a partially permeable membrane in dialysis and Visking tubing

Answer 3 of 7 · Question 5 part b · 3 marks

2025 JUN · Paper 01 (regional R) · extract 2 of 2 · find part b in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain diffusion across a partially permeable membrane in dialysis and Visking tubing

Answer 4 of 7 · Question 5 part a.iv · 1 mark

2025 JUN · Paper 01 (regional R) · extract 1 of 2 · find part a.iv in this mark scheme extract

5 (a)	(i)	• flows from artery; • where blood is at high pressure; • into vein; • because blood is at a low pressure;	mp1- accept in context of artery leaving heart mp3- accept in context vein of returning to the heart If comparative answer given for pressure score marking point 2 and 4	4
	(ii)	glomerulus;		1
	(iii)	• large (surface) area; • for diffusion / molecules/ named molecules to pass out(of blood);	More area for diffusion = 2 marks	2
	(iv)	more urea;		1
(b)		• no/less gradient between liquid X and blood (glucose)/blood and X have same concentration/eq; • glucose does not pass out/ glucose is not lost: • by diffusion • needed for respiration; • minimises osmotic effect;		3
(c)		• protein digested/broken down into <u>amino acids</u>; • (excess) **<u>amino acids</u>** broken down/deaminated; • urea; • less protein means less urea produced; • less urea means less to be filtered/removed ; • high concentration of urea/nitrogenous waste can be toxic;	MP3– accept urea from protein breakdown	4
				Total =15

Type MS · Explain diffusion across a partially permeable membrane in dialysis and Visking tubing

Answer 4 of 7 · Question 5 part a.iv · 1 mark

2025 JUN · Paper 01 (regional R) · extract 2 of 2 · find part a.iv in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain diffusion across a partially permeable membrane in dialysis and Visking tubing

Answer 5 of 7 · Question 2 part b · 5 marks

2022 JAN · Paper 01 · extract 1 of 2 · find part b in this mark scheme extract

Number				
2	(a) (i)	• sample placed in test tube; • Benedict's solution added; • heat in a water bath; • blue to brick red if glucose present/stays blue if absent;		4
	(ii)	• sample from tubing placed in test tube/on spotting tile; • iodine solution added; • brown to blue-black if present/stays brown if absent;		3
	(iii)	• remove any starch/glucose; • spilt when putting solution into tubing; • to avoid contamination/doesn't affect result/investigation/ensure results are valid;		3
	(b)	• water in tube, positive for glucose but negative for starch • contents of tubing should test positive for starch and glucose; • Visking tubing has pores/partially permeable; • size(of pores) allows glucose to pass; • starch can't pass;		5

Total : 15 marks

Type MS · Explain diffusion across a partially permeable membrane in dialysis and Visking tubing

Answer 5 of 7 · Question 2 part b · 5 marks

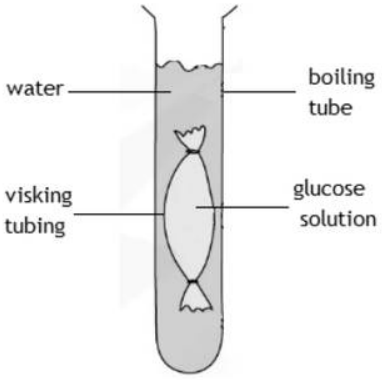
2022 JAN · Paper 01 · extract 2 of 2 · find part b in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain diffusion across a partially permeable membrane in dialysis and Visking tubing

Answer 6 of 7 · Question 5 part b.iii · 3 marks

2025 JUN · Paper 02 (regional R) · find part b.iii in this mark scheme extract

5 (a)		• greater concentration gradient/high concentration in small intestine; • (more) glucose collides/in contact with, wall/epithelium, of small intestine; • faster rate of diffusion;	allow reverse argument	3
(b)	(i)	Benedict's (solution);		1
	(ii)	(heated) water bath/Bunsen burner/measuring cylinder;		1
	(iii)	• diagram; • labels;;  	1 mark for each set of two correct labels. ignore any inclusion of a water bath in diagram accept sugar solution and glucose/sugar solution	3
(c)`	(i)	villus/villi		1
	(ii)	any three from • many/large number of villi/have microvilli; • increase surface area; • thin walls; • for shorter diffusion distance; • capillaries; • to maintain diffusion/concentration gradient;	explanation must be linked to feature	3
				Total for Question 5 = 12 marks

Type MS · Explain diffusion across a partially permeable membrane in dialysis and Visking tubing

Answer 7 of 7 · Question 2 part b.ii · 1 mark

2022 JUN · Paper 01 (regional R) · extract 1 of 2 · find part b.ii in this mark scheme extract

Number				
2	(a) (i)	W large intestine; X oesophagus; Y stomach; Z small intestine;	Allow colon Allow ileum	4
	(ii)	In the following order only: • bolus; • muscles; • contract; • peristalsis;		4
	(b) (i)	salivary gland/ pancreas;		1
	(ii)	(lining of) **small** intestine/ ileum/ duodenum;		1
	(iii)	• starch digested/broken down to sugar/glucose; • (sugar) diffuses/moves out through (Visking tubing);	Allow maltose	2
	(iv)	Benedict's solution/reagent;		1
	(v)	Any two from the following: • blue/black colour; • (undigested) starch is present; • too large to pass through Visking tubing;		2

Total marks 15

Type MS · Explain diffusion across a partially permeable membrane in dialysis and Visking tubing

Answer 7 of 7 · Question 2 part b.ii · 1 mark

2022 JUN · Paper 01 (regional R) · extract 2 of 2 · find part b.ii in this mark scheme extract

Question number	Answer	Notes	Marks